BRUNVOLL

Technical Report

Voyage Fuel Optimiser – Software Architecture & Validation

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1 Introduction

This report documents the architecture and validation of the `voyage_comparison` propulsion optimiser tool. The tool performs long-term hindcast-based voyage simulations comparing the factory combinator pitch/RPM schedule against a fuel-optimal operating point selection, with optional Flettner rotor wind-assist.

The tool was originally implemented as a single 4 913-line Python script. A modular refactoring was carried out to improve maintainability, testability, and extensibility. This report describes the physical models, the software architecture after refactoring, and presents validation results from a full-year simulation on the Rotterdam–Gothenburg route.

1.1 Scope

The tool addresses the following question: *for a given vessel on a fixed route, how much fuel can be saved by optimising the propeller pitch and RPM at each operating point, compared to the factory-default combinator schedule?*

The analysis chain covers:

- Calm-water hull resistance from service prediction data.
- Wave added resistance via PdStrip drift transfer functions integrated over JONSWAP spectra.
- Blendermann (1994) wind resistance model.
- Flettner rotor wind-assist thrust (Bordogna et al., 2019).
- Hull and propeller roughness penalties (Townsin–Dey, 2003).
- Factory combinator schedule replication.
- Fuel-optimal pitch/RPM selection via propeller model optimisation.

Hindcast weather is provided by NORA3 (Norwegian Reanalysis Archive, 3 km resolution) wave and wind fields.

2 Physical Models

2.1 Hull Resistance

Calm-water thrust demand is interpolated from the vessel 206 service prediction at 16 speeds ($8\,\mathrm{kn}$ to $15.5\,\mathrm{kn}$). The service prediction includes a 15% sea margin which is removed to obtain calm-water baseline values:

$$T_{\text{calm}}(V_s) = \frac{T_{\text{service}}(V_s)}{1.15} \tag{1}$$

Wake fraction $w(V_s)$ and thrust deduction $t(V_s)$ are also interpolated from service prediction tables.

2.2 Wave Added Resistance

Wave added resistance is computed from PdStrip strip-theory transfer functions. The `DriftTransferFunction` class loads the `.dat` output from PdStrip and computes the mean surge drift force for a given sea state:

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$$R_{\text{aw}} = 2\int_0^\infty S_\zeta(\omega;\, H_s, T_p)\cdot C_{\text{drift}}(\omega,\, \theta_{\text{rel}})\, d\omega \tag{2}$$

where S_ζ is the JONSWAP wave spectrum and C_{drift} is the frequency- and heading-dependent drift force transfer function per unit wave amplitude squared.

2.3 Wind Resistance

Hull wind resistance follows Blendermann (1994), using tabulated surge force coefficients $C_X(\alpha)$ for a general cargo vessel:

$$R_{\text{wind}} = -C_X(\alpha_{\text{app}})\cdot \tfrac{1}{2}\,\rho_{\text{air}}\, V_{\text{app}}^2 \cdot A_F \tag{3}$$

where α_{app} is the apparent wind angle from the bow, V_{app} the apparent wind speed, and $A_F = 280\,\mathrm{m}^2$ the frontal projected wind area.

2.4 Flettner Rotor

The Flettner rotor model is ported from the BruCon C++ vessel simulator. Lift and drag coefficients follow Bordogna et al. (2019) with aspect-ratio correction for a rotor of height $28\,\mathrm{m}$ and diameter $4\,\mathrm{m}$:

- Lift and drag are tabulated as functions of spin ratio SR $= \Omega R/V_{\text{app}}$.
- The target spin ratio is SR $= 3$ (max $220\,\mathrm{RPM}$).
- Rotor drive power follows the Theodorsen & Regier (1944) model.
- Rotor motor fuel is charged at the auxiliary genset SFOC ($215\,\mathrm{g}\cdot\mathrm{h/kW}$).

The net propulsive benefit is the surge component of the Magnus-effect force minus the fuel penalty for driving the rotor.

2.5 Hull and Propeller Roughness

Hull roughness resistance follows Townsin & Dey (2003):

$$\Delta C_F = 0.044\left[\left(\frac{k_s}{L}\right)^{1/3} - 10\,\text{Re}^{-1/3}\right] + 0.000125 \tag{4}$$

Propeller blade roughness increases torque (and thus fuel) via the empirical model of Mosaad (1986). Both roughness values can be specified directly in μm or via a `FoulingScenario` which models piecewise-linear roughness growth over time since last cleaning.

2.6 Factory Combinator

The `FactoryCombinator` class replicates the BruCon vessel simulator's `MakeNormalCombinator()` algorithm. For a given thrust demand, it returns the pitch and RPM dictated by the factory schedule—following the propeller curve up to a break point, then constant RPM with increasing pitch.

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2.7 Propeller Model and Fuel Optimisation

The fuel-optimal operating point is found by searching the full $(P/D,\, n)$ space of the C-series propeller model. For each thrust demand, the optimiser finds the pitch/RPM combination that minimises engine SFOC $\times$ brake power, subject to the MAN L27/38 power envelope.

Results are pre-computed for a grid of 681 thrust values ($10\,\mathrm{kN}$ to $350\,\mathrm{kN}$ in $0.5\,\mathrm{kN}$ steps) using parallel workers, then looked up during the voyage simulation for speed.

3 Software Architecture

3.1 Overview

The original implementation was a single file (`voyage_comparison.py`, 4 913 lines). The refactoring decomposed it into four sub-packages with a thin CLI entry point:

```
voyage_comparison.py       323 lines   CLI entry point (argparse)
models/                  1,500 lines   Physical models & data
simulation/              1,117 lines   Simulation engine & orchestration
plotting/                1,218 lines   Matplotlib visualisations
reporting/               1,023 lines   Text-based summary reports
                         -----
Total                    5,181 lines
```

3.2 `models/` Package

Contains all physical models, vessel data, and environmental data access.

Module	Lines	Description
`constants.py`	129	File paths, propeller/engine parameters, hull data arrays, Blendermann coefficient tables
`roughness.py`	268	`FoulingScenario` dataclass, Townsin–Dey hull roughness, blade roughness factor
`combinator.py`	256	`FactoryCombinator` – factory pitch/RPM schedule
`drift_force.py`	203	`DriftTransferFunction` – PdStrip wave added resistance
`flettner.py`	176	`FlettnerRotor` – Magnus-effect wind-assist model
`route.py`	168	`Waypoint`, `Route` – great-circle navigation
`weather.py`	150	`WeatherAlongRoute` – NORA3 NetCDF data loading
`wind_resistance.py`	108	Blendermann (1994) hull wind resistance
`geometry.py`	17	`angle_diff()` helper shared across models
`__init__.py`	35	Re-exports for convenient importing

3.3 `simulation/` Package

Contains the core simulation loop, caching infrastructure, and top-level orchestration functions.

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Module	Lines	Description
`engine.py`	445	`evaluate_voyage()` – hourly stepping loop, round-trip combiner, parallel worker
`orchestrator.py`	425	`run_annual_comparison()`, `run_speed_sweep()`, `run_scheduling_analysis()`
`cache.py`	131	Pre-computed thrust-to-fuel lookup tables (parallel build)
`results.py`	108	`HourlyResult`, `VoyageResult`, `SpeedSweepResult` dataclasses
`__init__.py`	13	Re-exports

The core simulation loop in `evaluate_voyage()` proceeds as follows:

1. Interpolate the route at 1-hour intervals (great-circle).
2. Compute speed-dependent quantities once: wake fraction, thrust deduction, calm-water thrust, hull roughness resistance, blade roughness factor.
3. For each hourly position:
   - Fetch NORA3 weather (H_s, T_p, wave/wind direction and speed).
   - Compute wave added resistance via drift transfer function.
   - Compute Flettner rotor thrust and power.
   - Compute Blendermann wind resistance.
   - Assemble total thrust demand: $T = T_{\text{calm}} + (R_{\text{aw}} + R_{\text{wind}} + R_{\text{rough}} - F_{\text{Flettner}})/(1-t)$.
   - Evaluate factory combinator and optimiser fuel rates (from cache).
   - Also evaluate no-Flettner variants for savings decomposition.
4. Aggregate fuel totals, weather means, and compute savings split.

Only hours where all four evaluations (factory $\pm$ Flettner, optimiser $\pm$ Flettner) are feasible are counted, ensuring a consistent comparison basis.

3.4 `plotting/` Package

Module	Lines	Description
`comparison.py`	481	Annual comparison time-series and histogram plots
`departure.py`	192	Departure timing flexibility analysis plots
`scheduling.py`	191	2D speed $\times$ departure window heatmaps
`map_plot.py`	189	Route map with Natural Earth coastlines
`speed_sweep.py`	163	Speed sensitivity sweep plots
`__init__.py`	8	Re-exports

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3.5 `reporting/` Package

Module	Lines	Description
`summary.py`	519	`print_summary()`, `print_departure_analysis()`
`scheduling.py`	390	`print_scheduling_analysis()`, `print_roughness_sweep_summary()`
`speed_sweep.py`	106	`print_speed_sweep_summary()`
`__init__.py`	12	Re-exports

3.6 CLI Interface

The entry point `voyage_comparison.py` exposes 29 command-line arguments via `argparse`. Key modes of operation:

Flag	Description
`--download`	Pre-download NORA3 data for the route
`--year YEAR`	Hindcast year (default: 2024)
`--speed SPEED`	Transit speed in knots (default: 10)
`--plot`	Generate summary plots
`--no-flettner`	Disable Flettner rotor
`--compare-sg`	Standard vs shaft generator comparison
`--speed-sweep KN ...`	Speed sensitivity sweep
`--roughness-sweep T ...`	Fouling age sensitivity sweep
`--departure-analysis`	Departure timing flexibility analysis
`--scheduling-analysis KN ...`	2D departure $\times$ speed optimisation
`--hull-roughness KS_UM`	Hull roughness k_s [\$$\mu$\$m]
`--fouling-years T`	Time since cleaning [years]

4 Validation Results

A full-year simulation was run on the Rotterdam–Gothenburg route for 2024 as an end-to-end validation of the refactored codebase:

```
python voyage_comparison.py --year 2024 --quiet
```

4.1 Weather Data

The simulation uses wave and wind data from NORA3, the third-generation Norwegian Reanalysis Archive produced by MET Norway. A *hindcast* is a numerical weather simulation run retrospectively: the atmospheric and wave models are forced with observed boundary conditions and assimilate available measurements, producing a physically consistent, gap-free reconstruction of historical weather on a regular grid. Unlike point measurements from buoys or ships, a hindcast provides complete spatial and temporal coverage, making it suitable for route-based voyage simulation.

NORA3 uses the WAM spectral wave model driven by the HARMONIE-AROME atmospheric model. The wave data used here covers the North Sea and Skagerrak on a rotated-pole grid with approximately $3\,\mathrm{km}$ spatial resolution (Δlat $\approx 0.02°$, Δlon $\approx 0.035°$, corresponding to $\approx 2.2\,\mathrm{km} \times 2.2\,\mathrm{km}$ at the route latitude) and hourly temporal resolution. The data spans 1993 to the present.

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The variables extracted for each route point are:

- H_s – significant wave height [m]
- T_p – peak wave period [s]
- θ_q – mean wave direction [deg, meteorological convention]
- U_{10} – 10 m wind speed [m/s]
- θ_w – wind direction [deg, meteorological convention]

Monthly files are pre-downloaded for a bounding box around the route ($\pm 0.5^\circ$ margin) and stored as NetCDF files. During simulation, the weather at each hourly vessel position is obtained by nearest-neighbour lookup in both space and time. Given the $3\,\mathrm{km}$ grid spacing, the maximum spatial interpolation error is approximately $1.5\,\mathrm{km}$ – well within the spatial decorrelation scale of the wave field. The hourly time step matches the simulation stepping interval exactly, so temporal interpolation is not required.

4.2 Simulation Setup

Parameter	Value	Unit
Route	509	nm (one-way)
Transit speed	10	kn
Transit time (one-way)	50.9	hours
Mode	Round-trip	
Flettner rotor	Enabled	
Hull roughness	Clean (no fouling)	
Blade roughness	Clean (no fouling)	
Idle/port time	15	%
Fuel price	650	EUR/tonne
Propeller	$D = 4.66$	m
Engine	MAN L27/38	8-cyl, 800 RPM

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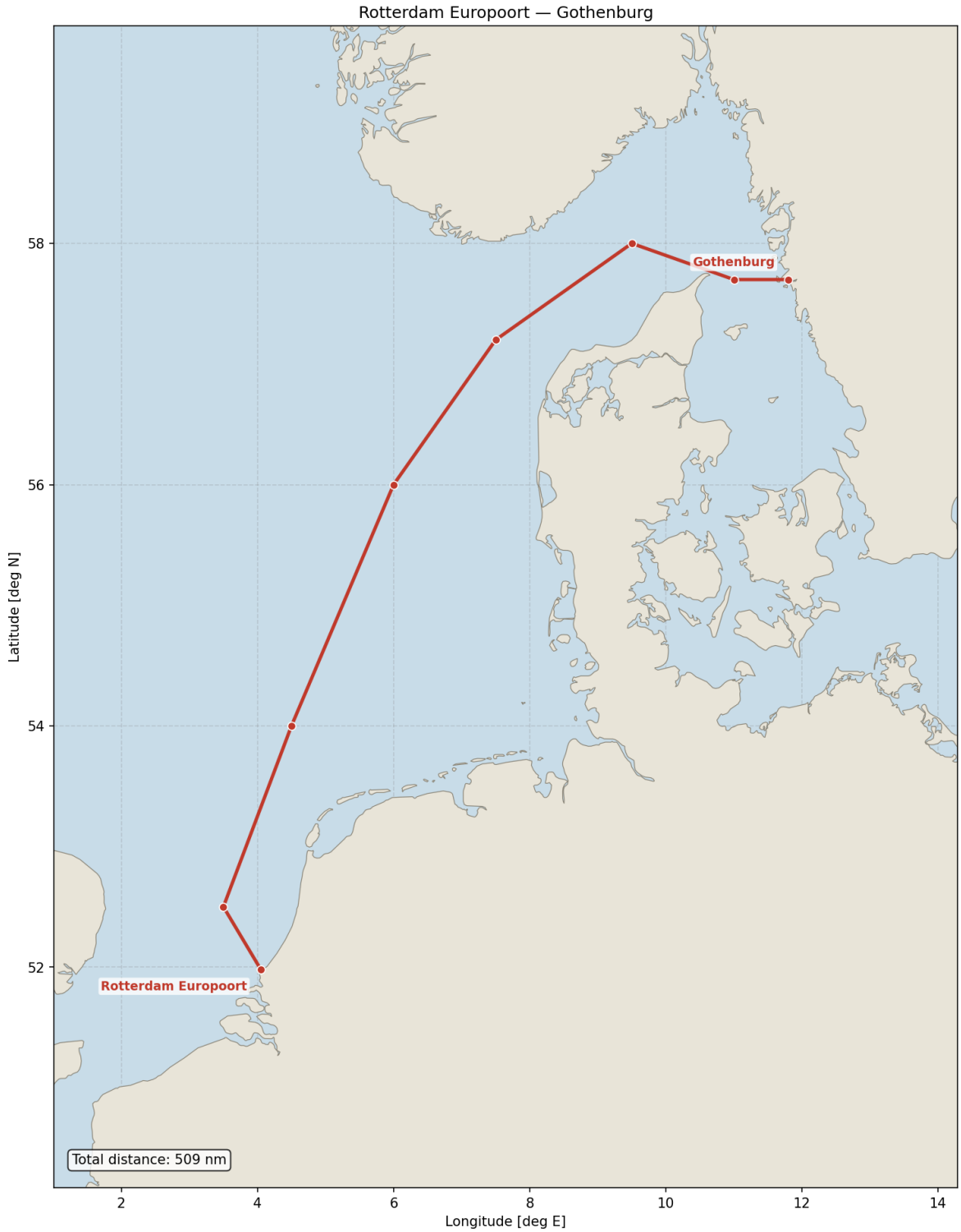

Figure 4-1: Rotterdam–Gothenburg route (509 nm one-way) with waypoints and leg bearings. Coastline data from Natural Earth 10m.

4.3 Voyage Statistics

362 round-trip voyages were successfully simulated (4 late-December departures failed due to missing January 2025 weather data).

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Metric	Mean	Median	P10	P90	Min	Max
Factory no-Fl [kg/voy]	14110	13981	12900	15806	10315	18946
Opt + Flettner [kg/voy]	12517	12557	10479	14385	7967	18014
Pitch/RPM saving [kg/voy]	415	431	335	473	183	497
Flettner saving [kg/voy]	1178	1084	309	2239	-3	3687
Pitch/RPM saving	3.0%	3.1%	2.2%	3.6%	1.3%	3.9%
Flettner saving	8.5%	7.6%	2.2%	16.2%	0.0%	27.4%
Mean H_s [m]	1.4	1.3	0.7	2.3	0.4	4.0
Mean wind speed [m/s]	7.8	7.6	5.3	10.6	3.5	14.5
Mean R_{aw} [kN]	4.7	2.1	0.2	12.4	-0.9	45.2
Mean R_{wind} [kN]	4.5	4.0	-0.6	10.5	-4.5	26.9
Mean Flettner thrust [kN]	14.3	11.5	3.2	30.5	0.1	55.7
Mean rotor power [kW]	17.4	13.9	4.0	37.6	0.2	68.0

Fuel savings are decomposed additively against the factory-no-Flettner baseline:

$$\underbrace{\text{Factory}_{\text{NF}} - \text{Opt}_{+\text{Fl}}}_{\text{combined saving}} = \underbrace{\text{Factory}_{\text{NF}} - \text{Opt}_{\text{NF}}}_{\text{pitch/RPM}} + \underbrace{\text{Opt}_{\text{NF}} - \text{Opt}_{+\text{Fl}}}_{\text{Flettner}} \tag{5}$$

The pitch/RPM saving (3.0%) isolates the benefit of optimising the operating point at the same thrust demand. The Flettner saving (8.5%) isolates the thrust reduction from the rotor. Both percentages are relative to the factory-no-Flettner baseline.

Figure 4-2 shows the daily evolution of fuel savings, significant wave height, and force components across the full year.

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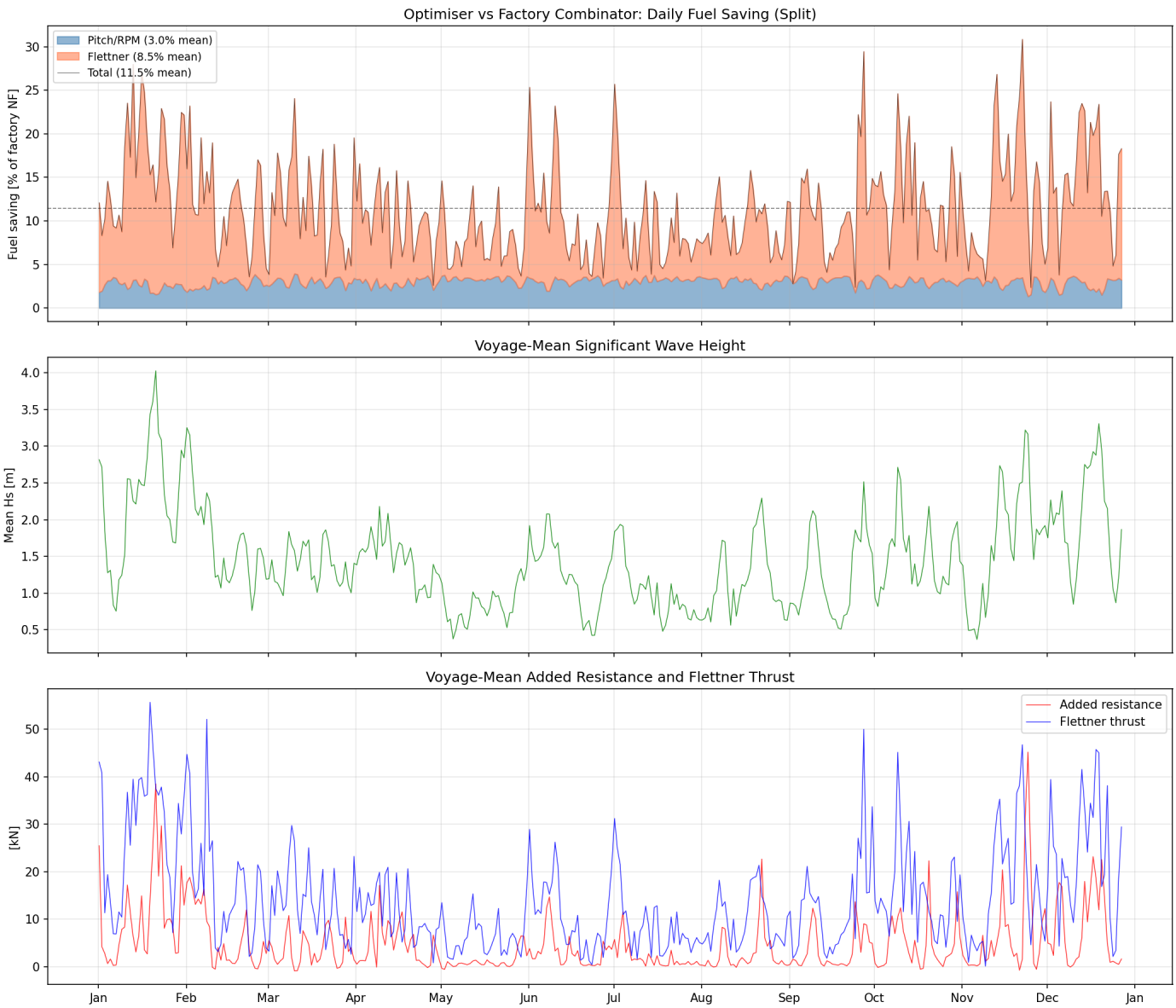

Figure 4-2: Annual time-series of per-voyage results. Top: pitch/RPM and Flettner fuel savings (stacked). Middle: significant wave height. Bottom: added wave resistance, wind resistance, and Flettner thrust.

4.4 Factory Combinator Feasibility

Metric	Value
Total evaluation hours	37 648
Both feasible	97%
Factory infeasible hours	1%
Voyages with ≥ 1 infeasible hour	7%

Factory infeasibility occurs when the engine exceeds its power envelope at the factory combinator's fixed RPM schedule. The optimiser adapts pitch/RPM to remain feasible.

Figure 4-3 shows the correlation between per-voyage fuel saving and mean sea state, coloured by mean wind speed. Higher seas drive larger absolute savings through the Flettner rotor.

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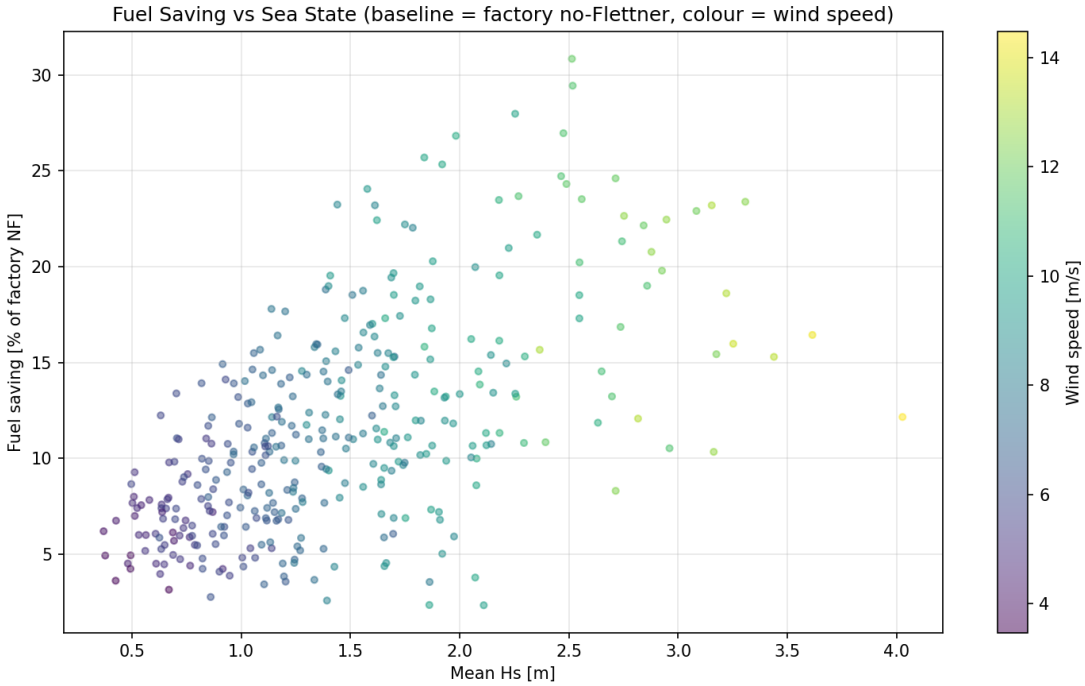

Figure 4-3: Per-voyage fuel saving vs mean significant wave height, coloured by mean wind speed. The positive correlation reflects the Flettner rotor's increasing benefit in windier conditions.

4.5 Annualised Fuel Consumption

Assuming 15% idle time, the vessel completes approximately 73 round-trip voyages per year.

Configuration	Fuel [tonnes/year]
Factory, no Flettner	1031.8
Factory + Flettner	945.1
Optimiser, no Flettner	1001.4
Optimiser + Flettner	915.3

Saving component	Tonnes/year	Percentage
Pitch/RPM	30.3	2.9%
Flettner	86.1	8.3%
Combined	116.4	11.3%

Figure 4-4 shows the distribution of per-voyage savings, decomposed into pitch/RPM optimisation and Flettner wind-assist components.

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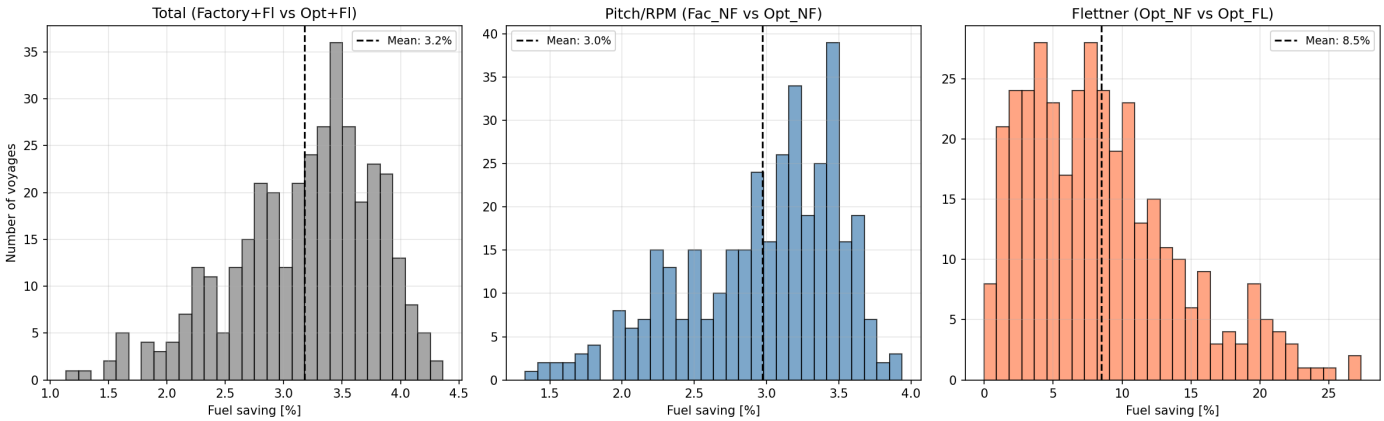

Figure 4-4: Histograms of per-voyage fuel savings. Left: total saving (factory vs optimiser+Flettner). Centre: pitch/RPM component. Right: Flettner component. The Flettner saving has a wider spread reflecting its weather dependence.

4.6 Cost Estimates

At MGO price of €650/tonne (typical ECA zone):

Configuration	Annual cost
Factory, no Flettner	€670 656
Factory + Flettner	€614 328
Optimiser, no Flettner	€650 940
Optimiser + Flettner	€594 964
Annual saving	€75 692 (11.3%)

4.7 Seasonal Breakdown

Season	Voyages	Fac NF [kg]	Opt+Fl [kg]	Saving	P/R	Flettner	Mean H_s [m]
Q1 (Jan–Mar)	91	14 102	12 266	3.0%	2.8%	10.4%	1.8
Q2 (Apr–Jun)	91	14 222	12 876	3.3%	3.1%	6.5%	1.2
Q3 (Jul–Sep)	92	14 068	12 711	3.3%	3.1%	6.6%	1.1
Q4 (Oct–Dec)	88	14 046	12 204	3.0%	2.9%	10.5%	1.7

The Flettner benefit is roughly doubled in the winter quarters (Q1, Q4) compared to summer, reflecting the stronger and more consistent North Sea wind regime. The pitch/RPM optimisation saving is relatively stable across seasons at 2.8–3.1%, as it depends primarily on the thrust demand distribution rather than its magnitude.

5 Conclusions

The refactored `voyage_comparison` tool has been validated by running a full-year (2024) hindcast simulation on the Rotterdam–Gothenburg route. Key findings:

1. **Pitch/RPM optimisation** saves approximately **2.9% fuel** (30 tonnes/year, €20 000/year) compared to the factory combinator schedule. This is a conservative, weather-independent benefit available on every voyage.

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2. **Flettner rotor wind-assist** adds a further **8.3% fuel saving** (86 tonnes/year, €56 000/year), with strong seasonal variation (6–10% by quarter).
3. **Combined saving** of **11.3%** (116 tonnes/year, €76 000/year) at $10\,\mathrm{kn}$ transit speed.
4. The factory combinator becomes **infeasible** (engine over-power) on 7% of voyages, while the optimiser adapts pitch/RPM to remain within the power envelope.
5. The **modular architecture** (4 packages, 24 modules, 5 181 lines) provides clean separation of physical models, simulation logic, reporting, and visualisation, enabling independent testing and extension of each component.

6 References

- Blendermann, W. (1994). Parameter identification of wind loads on ships. *J. Wind Eng. Ind. Aerodyn.*, 51(3), 339–351.
- Bordogna, G., Muggiasca, S., Giappino, S., Belloli, M., Keuning, J.A., Huijsmans, R.H.J. (2019). The effects of the aerodynamic interaction on the performance of two Flettner rotors. *J. Wind Eng. Ind. Aerodyn.*, 196, 104024.
- Mosaad, M.A. (1986). Marine propeller roughness penalties. PhD Thesis, University of Newcastle upon Tyne.
- Theodorsen, T. & Regier, A. (1944). Experiments on drag of revolving disks, cylinders, and streamline rods at high speeds. *NACA Report 793*.
- Townsin, R.L. & Dey, S.K. (2003). The correlation of roughness drag with surface characteristics. In *Biofouling*, Royal Institution of Naval Architects.

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